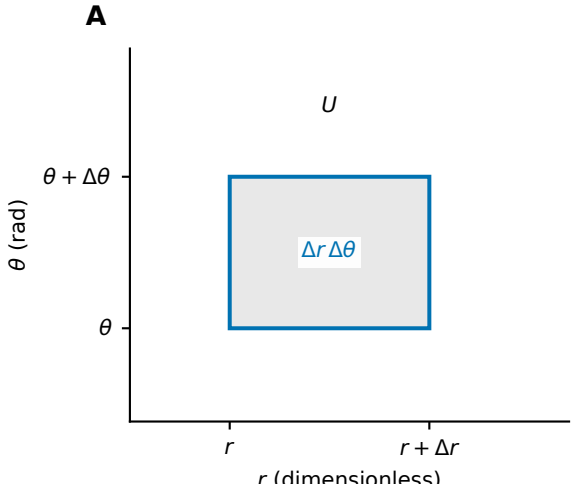
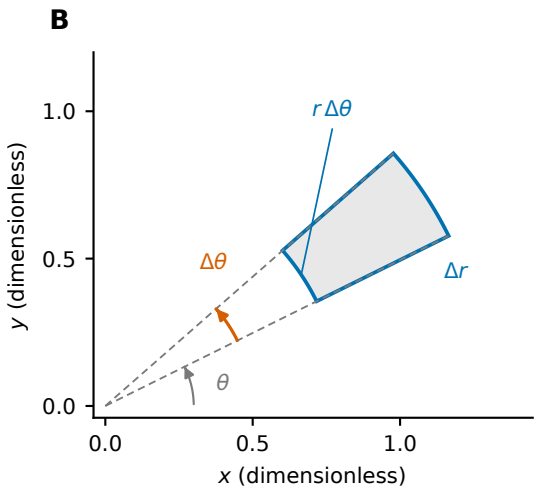


$T(r, \theta) = (r\cos\theta, r\sin\theta)$



$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$